

HDOC – 13

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Subject – C Programming

❖ PROBLEM STATEMENT:

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Count the number of even digits, and odd digits.
- Find the sum of its even digits and the sum of its odd digits separately.

○ ALGORITHM:

- 1) Start
- 2) Declare variables num, digit, choice, even_digit, odd_digit, sum_even, sum_odd as integer.
- 3) Read variable num.
- 4) Display the menu:
 - Choice 1: Count even and odd digits.
 - Choice 2: Find the sum of even and odd digits
- 5) Read choice.
- 6) Use switch-case.
- 7) If choice is 1:
 - Take the last digit using $\text{num \% 10}$.
 - Check whether the digit is even or odd.
 - Increase even_digit or odd_digit.
 - Remove the last digit using num / 10 .

Repeat until num becomes 0.

Display the count of even and odd digits.

8) If choice is 2:

Take the last digit using $\text{num} \% 10$.

If the digit is even, add it to `sum_even`.

Otherwise, add it to `sum_odd`.

Remove the last digit using $\text{num} / 10$.

Repeat until num becomes 0.

Display the sums, go to step 10.

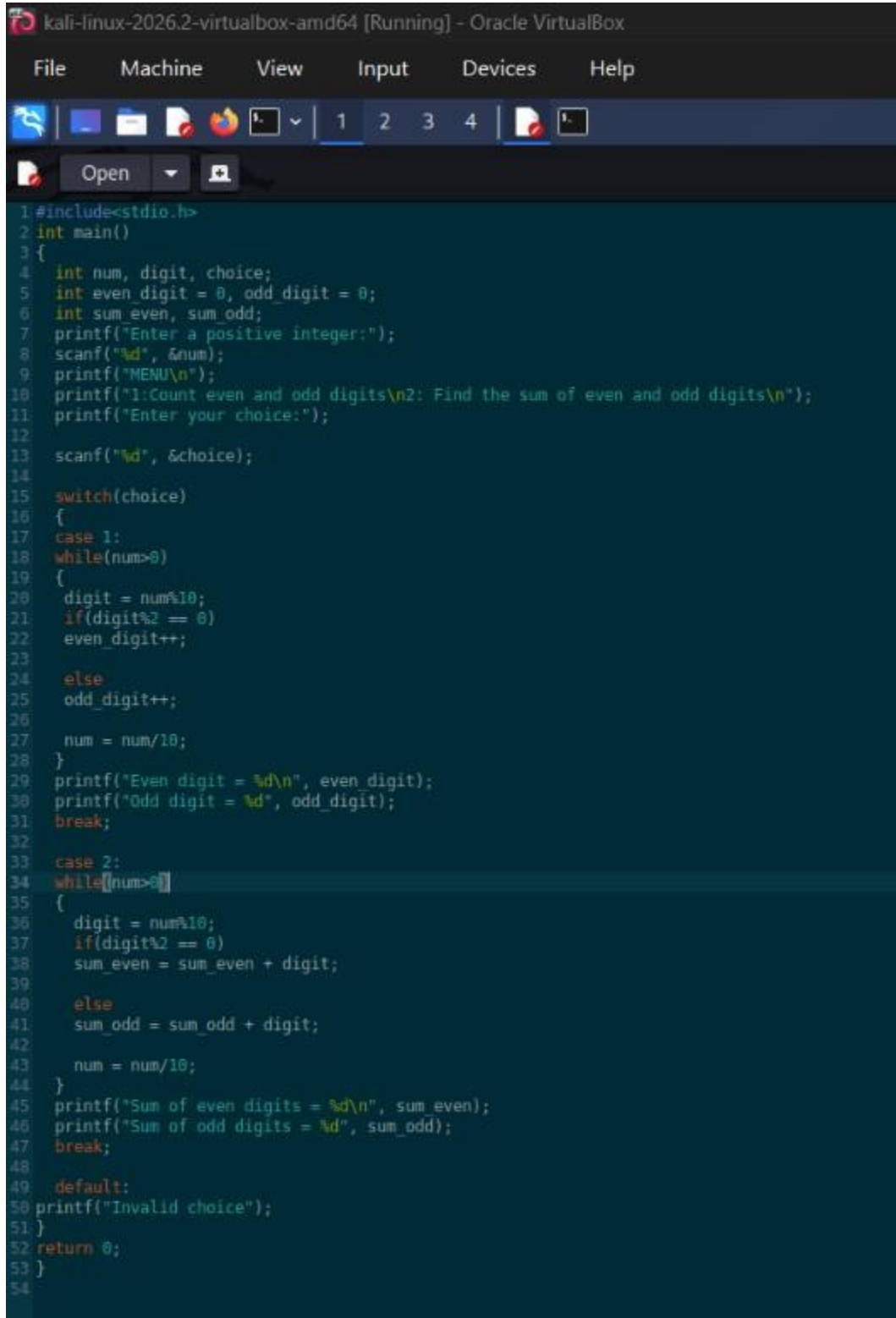
9) If the choice is invalid, display invalid choice.

10) Stop

○ TEST CASE TABLE:

Test case	Input	Choice	Expected output	Actual output	Result
1.	4739812	1	Even digits =3 Odd digits = 4	Even digits =3 Odd digits =4	Pass
2.	5438932	2	Sum of even digits=14 Sum of odd digits=20	Sum of even digits=14 Sum of odd digits=20	Pass
3.	7502372	1	Even digits =3 Odd digits = 4	Even digits =3 Odd digits =4	Pass
4.	4582402	2	Sum of even digits=20 Sum of odd digits=5	Sum of even digits=20 Sum of odd digits=5	Pass
5.	5768042	3	Invalid choice	Invalid choice	Pass

○ CODE:



```
1 #include<stdio.h>
2 int main()
3 {
4     int num, digit, choice;
5     int even_digit = 0, odd_digit = 0;
6     int sum_even, sum_odd;
7     printf("Enter a positive integer:");
8     scanf("%d", &num);
9     printf("MENU\n");
10    printf("1:Count even and odd digits\n2: Find the sum of even and odd digits\n");
11    printf("Enter your choice:");
12
13    scanf("%d", &choice);
14
15    switch(choice)
16    {
17        case 1:
18            while(num>0)
19            {
20                digit = num%10;
21                if(digit%2 == 0)
22                    even_digit++;
23
24                else
25                    odd_digit++;
26
27                num = num/10;
28            }
29            printf("Even digit = %d\n", even_digit);
30            printf("Odd digit = %d", odd_digit);
31            break;
32
33        case 2:
34            while(num>0)
35            {
36                digit = num%10;
37                if(digit%2 == 0)
38                    sum_even = sum_even + digit;
39
40                else
41                    sum_odd = sum_odd + digit;
42
43                num = num/10;
44            }
45            printf("Sum of even digits = %d\n", sum_even);
46            printf("Sum of odd digits = %d", sum_odd);
47            break;
48
49        default:
50            printf("Invalid choice");
51    }
52    return 0;
53 }
54
```

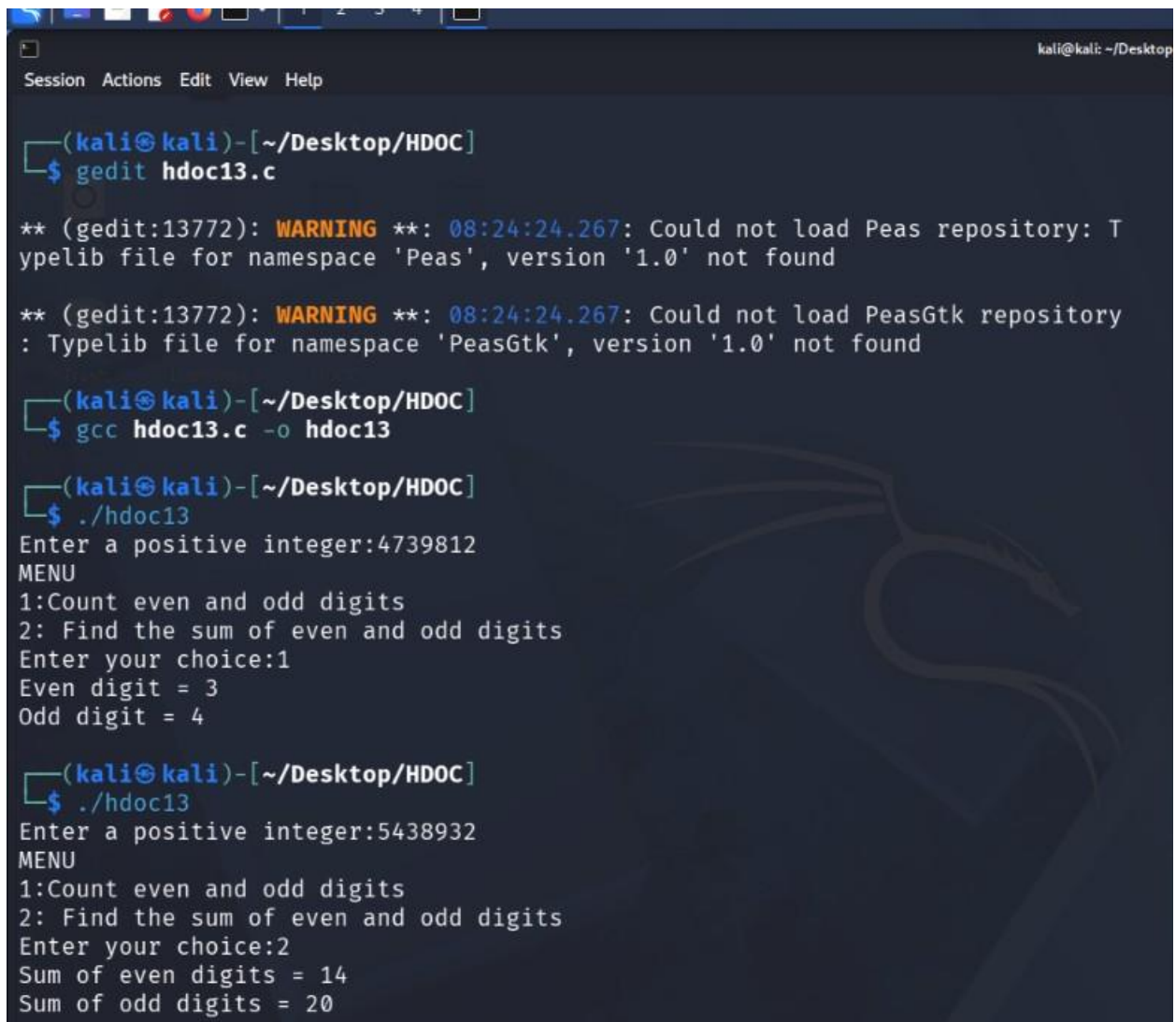
Compile the code using:

```
gcc hdoc13.c -o hdoc13
```

Execute the code using:

```
./hdoc13
```

Here are some screenshots of the compilation and execution:



```
(kali@kali)-[~/Desktop/HDOC]
$ gedit hdoc13.c

** (gedit:13772): WARNING **: 08:24:24.267: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:13772): WARNING **: 08:24:24.267: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

(kali@kali)-[~/Desktop/HDOC]
$ gcc hdoc13.c -o hdoc13

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc13
Enter a positive integer:4739812
MENU
1:Count even and odd digits
2: Find the sum of even and odd digits
Enter your choice:1
Even digit = 3
Odd digit = 4

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc13
Enter a positive integer:5438932
MENU
1:Count even and odd digits
2: Find the sum of even and odd digits
Enter your choice:2
Sum of even digits = 14
Sum of odd digits = 20
```

```
Session Actions Edit View Help
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc13
Enter a positive integer:7502372
MENU
1:Count even and odd digits
2: Find the sum of even and odd digits
Enter your choice:1
Even digit = 3
Odd digit = 4

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc13
Enter a positive integer:4582402
MENU
1:Count even and odd digits
2: Find the sum of even and odd digits
Enter your choice:2
Sum of even digits = 20
Sum of odd digits = 5

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc13
Enter a positive integer:5768042
MENU
1:Count even and odd digits
2: Find the sum of even and odd digits
Enter your choice:3
Invalid choice

(kali@kali)-[~/Desktop/HDOC]
$ gedit hdoc13.c
```